

WEEK – 5

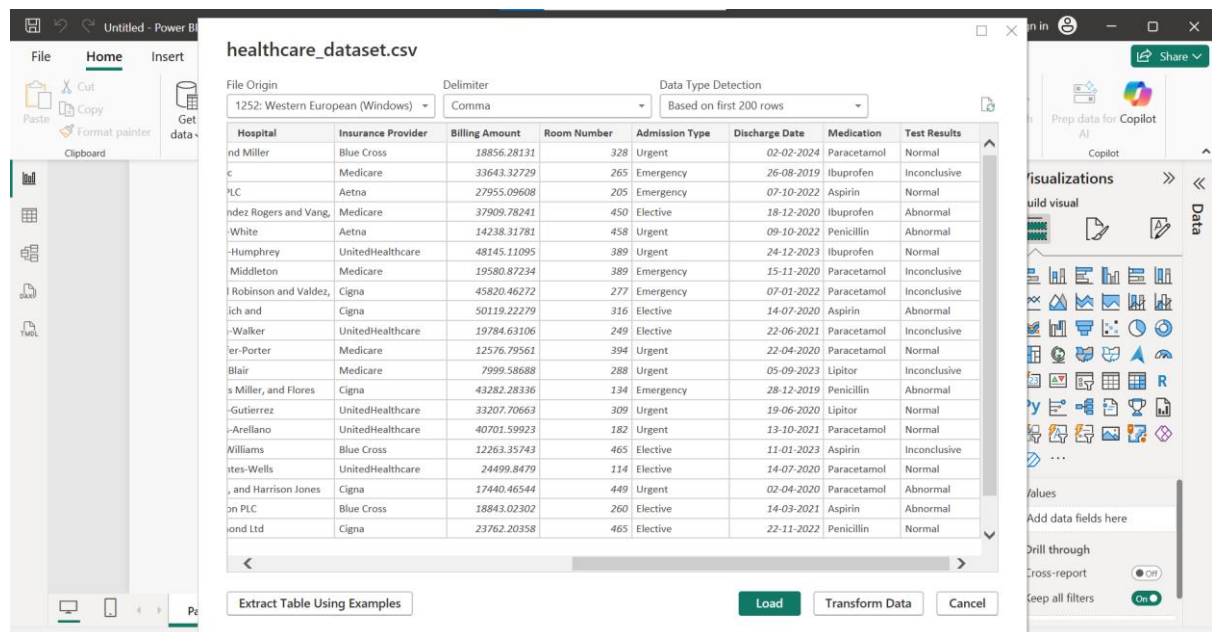
Healthcare Data Cleaning & Data Modeling using Power BI

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1. Introduction

This report presents the cleaning, transformation, and data modeling of a healthcare dataset using Power BI. The raw healthcare data was processed using Power Query and structured into dimension and fact tables for analysis.



Transforming the healthcare dataset into Power BI.

2. Data Cleaning – Trim & Clean

Removed extra spaces and unwanted non-printable characters from the text columns using **Trim** and **Clean** in Power Query.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Data Type: Text

Replace Values Fill

Unpivot Columns Move Convert to List

Any Column

Split Column

Format

Text Column

Merge Columns

Extract

Parse

Statistics Standard Scientific

10²

Trigonometry

Rounding

Information

Number Column

Date

Time

Duration

Run R script

Run Python script

Scripts

Queries [1]

healthcare_dataset

Room Number Admission Type Discharge Date Medication Test Results

Valid 100% Valid 100% Valid 100% Valid 100% Valid 100%

Error 0% Error 0% Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0% Empty 0% Empty 0%

1 328 Urgent 02-02-2024 Paracetamol Normal

2 265 Emergency 26-08-2019 Ibuprofen Inconclusive

3 205 Emergency 07-10-2022 Aspirin Normal

4 450 Elective 18-12-2020 Ibuprofen Abnormal

5 458 Urgent 09-10-2022 Penicillin Abnormal

6 389 Urgent 24-12-2023 Ibuprofen Normal

7 389 Emergency 15-11-2020 Paracetamol Inconclusive

8 277 Emergency 07-01-2022 Paracetamol Inconclusive

9 316 Elective 14-07-2020 Aspirin Abnormal

10 249 Elective 22-06-2021 Paracetamol Inconclusive

11 394 Urgent 22-04-2020 Paracetamol Normal

12 288 Urgent 05-09-2023 Lipitor Inconclusive

13 134 Emergency 28-12-2019 Penicillin Abnormal

14 309 Urgent 19-06-2020 Lipitor Normal

15 182 Urgent 13-10-2021 Paracetamol Normal

16 465 Elective 11-01-2023 Aspirin Inconclusive

17 114 Elective 14-07-2020 Paracetamol Normal

18 449 Urgent 02-04-2020 Paracetamol Abnormal

19

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 10:31

Query Settings

PROPERTIES

Name healthcare_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Trimmed Text

Cleaned Text

Applied Trim and Clean transformations to the healthcare data.

3. Adding Age Group Column

Created an **Age Group** column using the patient's age to categorize patients into different age groups.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function

General

Conditional Column

Index Column

Duplicate Column

Format

Parse

Merge Columns

Extract

Parse

Statistics Standard Scientific

10²

Trigonometry

Rounding

Information

Date

Time

Duration

Queries [1]

healthcare_dataset

Admission Type

Valid

Error

Empty

1 ent

2 rgency

3 rgency

4 tive

5 ent

6 ent

7 rgency

8 rgency

9 tive

10 tive

11 ent

12 ent

13 rgency

14 ent

15 ent

16 tive

17 tive

18 ent

19

16 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Custom Column

Add a column that is computed from the other columns.

New column name

Age Group

Custom column formula

= if[Age]<=18 then "0-18"
else if[Age]<=35 then "19-35"
else if[Age]<=60 then "36-60"
else "60+"

Available columns

Name

Age

Gender

Blood Type

Medical Condition

Date of Admission

Doctor

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

13-10-2021 Paracetamol Normal 36-60

11-01-2023 Aspirin Inconclusive 36-60

14-07-2020 Paracetamol Normal 60

02-04-2020 Paracetamol Abnormal 36-60

Query Settings

PROPERTIES

Name healthcare_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Trimmed Text

Cleaned Text

Added Conditional Column

Created an Age Group column using Power Query.

4. Creating and Renaming Patient ID

Created a unique **Patient ID** using an Index column and renamed the column to **Patient ID**.

The screenshot shows the Power Query Editor interface. The main area displays a table named 'Dim_Patient' with columns: Age, Gender, Blood Type, Age Group, and Patient ID. The Patient ID column is highlighted in green. The formula bar shows the transformation: `= Table.RenameColumns(#"Added Index",{{"Index", "Patient ID"}})`. The right sidebar shows the 'Query Settings' for 'Dim_Patient', with the 'Applied Steps' list including 'Renamed Columns'.

	Age	Gender	Blood Type	Age Group	Patient ID
1	30	Male	B-	19-35	1
2	62	Male	A+	60+	2
3	76	Female	A-	60+	3
4	28	Female	O+	19-35	4
5	43	Female	AB+	36-60	5
6	36	Male	A+	36-60	6
7	21	Female	AB-	19-35	7
8	20	Female	A+	19-35	8
9	82	Male	AB+	60+	9
10	58	Female	AB-	36-60	10
11	72	Male	O+	60+	11
12	38	Female	A-	36-60	12
13	75	Female	A+	60+	13
14	68	Female	AB+	60+	14
15	44	Female	AB+	36-60	15
16	46	Female	AB-	36-60	16
17	63	Female	A+	60+	17
18	38	Male	AB-	36-60	18

Created and renamed the Patient ID column in the Dim_Patient table.

5. Creating and Renaming Admission ID

Created a unique **Admission ID** using an Index column and renamed it to **Admission_ID**.

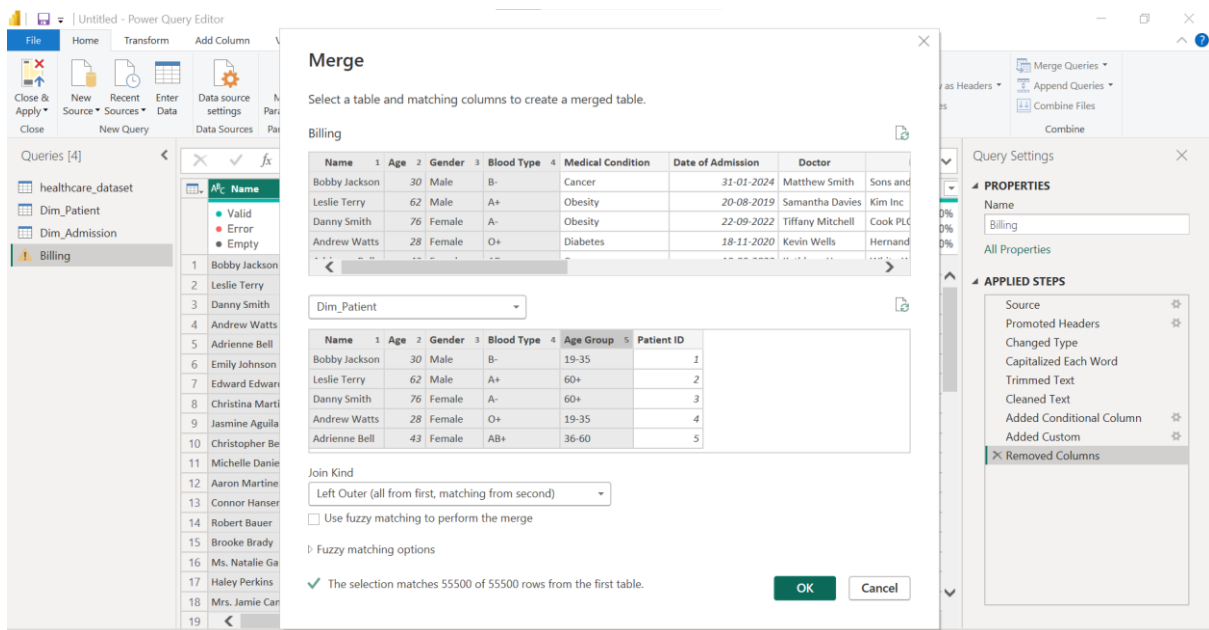
The screenshot shows the Power Query Editor interface. The main area displays a table named 'Dim_Admission' with columns: Medical Condition, Medication, Room Number, Test Results, and Admission_ID. The Admission_ID column is highlighted in green. The formula bar shows the transformation: `= Table.RenameColumns(#"Removed Columns1",{{"Index.1", "Admission_ID"}})`. The right sidebar shows the 'Query Settings' for 'Dim_Admission', with the 'Applied Steps' list including 'Renamed Columns'.

	Medical Condition	Medication	Room Number	Test Results	Admission_ID
1	cer	Paracetamol	328	Normal	1
2	sity	Ibuprofen	265	Inconclusive	2
3	sity	Aspirin	205	Normal	3
4	betes	Ibuprofen	450	Abnormal	4
5	cer	Penicillin	458	Abnormal	5
6	rma	Ibuprofen	389	Normal	6
7	betes	Paracetamol	389	Inconclusive	7
8	cer	Paracetamol	277	Inconclusive	8
9	rma	Aspirin	316	Abnormal	9
10	cer	Paracetamol	249	Inconclusive	10
11	cer	Paracetamol	394	Normal	11
12	ertension	Lipitor	288	Inconclusive	12
13	betes	Penicillin	134	Abnormal	13
14	rma	Lipitor	309	Normal	14
15	cer	Paracetamol	182	Normal	15
16	sity	Aspirin	465	Inconclusive	16
17	ritis	Paracetamol	114	Normal	17
18	sity	Paracetamol	449	Abnormal	18

Created and renamed the Admission ID column in the Dim_Admission table.

6. Merge Tables

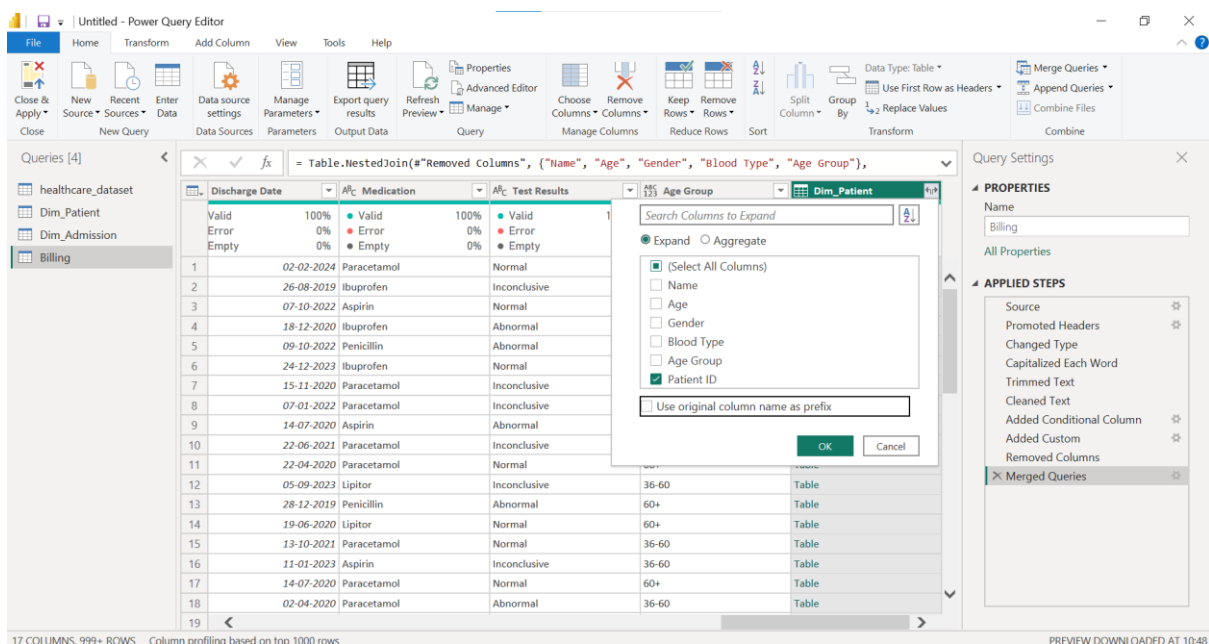
Merged the **Billing** table with the **Dim_Patient** table using the matching patient information to connect the data.



Merged Billing and Dim_Patient tables in Power Query.

7. Expand Dim_Patient

Expanded the merged **Dim_Patient** table and selected **Patient ID** to add the required patient information to the Billing table.



Expanded the merged Dim_Patient table and added Patient ID.

7. Expanded Patient ID

Expanded the merged Dim_Patient table and added the **Patient ID** column to the Billing table.

The screenshot shows the Power Query Editor interface. The main area displays a table with 17 columns and 999+ rows. The columns are: Discharge Date, Medication, Test Results, Age Group, and Patient ID. The 'Patient ID' column is highlighted in green. The table contains 18 rows of data. The 'Query Settings' pane on the right shows the 'Expanded Dim_Patient' step.

	Discharge Date	Medication	Test Results	Age Group	Patient ID
1	02-02-2024	Paracetamol	Normal	19-35	1
2	26-08-2019	Ibuprofen	Inconclusive	60+	2
3	07-10-2022	Aspirin	Normal	60+	3
4	18-12-2020	Ibuprofen	Abnormal	19-35	4
5	09-10-2022	Penicillin	Abnormal	36-60	5
6	24-12-2023	Ibuprofen	Normal	36-60	6
7	15-11-2020	Paracetamol	Inconclusive	19-35	7
8	07-01-2022	Paracetamol	Inconclusive	19-35	8
9	14-07-2020	Aspirin	Abnormal	60+	9
10	22-06-2021	Paracetamol	Inconclusive	36-60	10
11	22-04-2020	Paracetamol	Normal	60+	11
12	05-09-2023	Lipitor	Inconclusive	36-60	12
13	28-12-2019	Penicillin	Abnormal	60+	13
14	19-06-2020	Lipitor	Normal	60+	14
15	13-10-2021	Paracetamol	Normal	36-60	15
16	11-01-2023	Aspirin	Inconclusive	36-60	16
17	14-07-2020	Paracetamol	Normal	60+	17
18	02-04-2020	Paracetamol	Abnormal	36-60	18

Added Patient ID to the Billing table after expanding Dim_Patient.

8. Merge Dim_Admission

Merged the **Billing** table with the **Dim_Admission** table using matching admission details.

Merge

Select a table and matching columns to create a merged table.

Billing

Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group	Patient ID
31	328	Urgent	02-02-2024	Paracetamol	Normal	19-35
29	265	Emergency	26-08-2019	Ibuprofen	Inconclusive	60+
08	205	Emergency	07-10-2022	Aspirin	Normal	60+
41	450	Elective	18-12-2020	Ibuprofen	Abnormal	19-35

Dim_Admission

Hospital	Medical Condition	Medication	Room Number	Test Results	Admission_ID
Sons and Miller	Cancer	Paracetamol	328	Normal	1
Kim Inc	Obesity	Ibuprofen	265	Inconclusive	2
Cook PLC	Obesity	Aspirin	205	Normal	3
Hernandez Rogers and Vang,	Diabetes	Ibuprofen	450	Abnormal	4

Join Kind: Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

☒ The selection matches 56612 of 56612 rows from the first table.

OK Cancel

Query Settings

PROPERTIES

Name: Billing

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Added Conditional Column
- Added Custom
- Removed Columns
- Merged Queries
- Expanded Dim_Patient

Merged Billing with Dim_Admission to connect admission information.

9. Expand Dim_Admission

Expanded the merged Dim_Admission table and selected **Admission_ID** to add the admission information to the Billing table.

Expand

Search Columns to Expand

☒ Expand ☐ Aggregate

☒ (Select All Columns)

☐ Admission Type

☐ Doctor

☐ Hospital

☐ Medical Condition

☐ Medication

☐ Room Number

☐ Test Results

☒ Admission_ID

☐ Use original column name as prefix

OK Cancel

Query Settings

PROPERTIES

Name: Billing

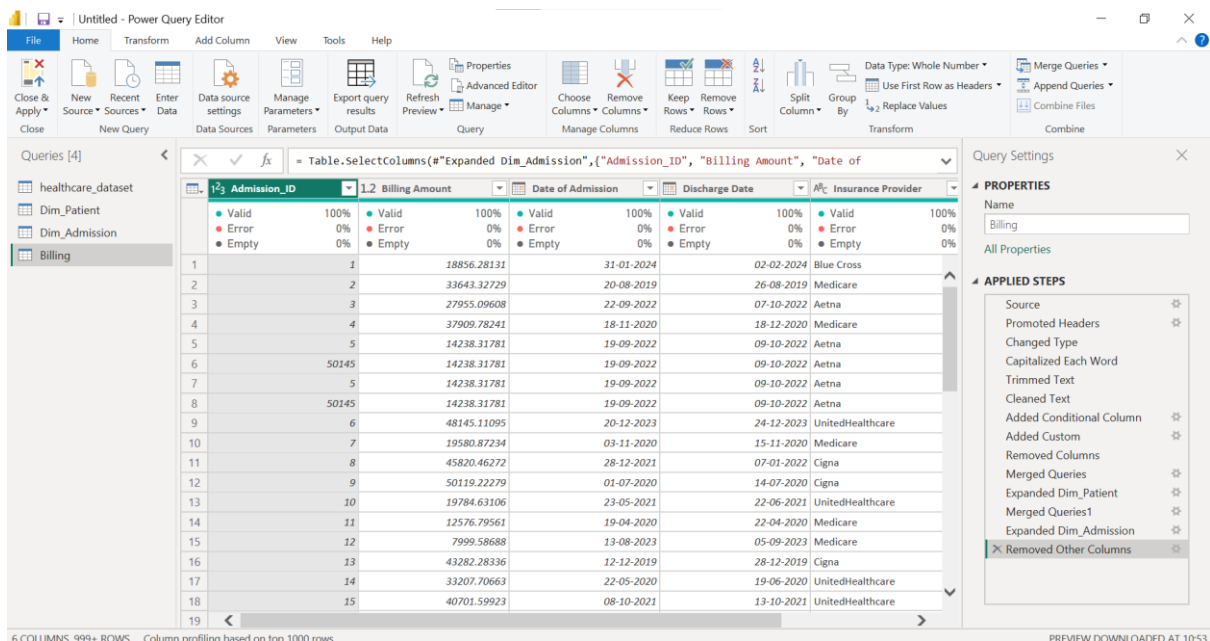
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Added Conditional Column
- Added Custom
- Removed Columns
- Merged Queries
- Expanded Dim_Patient
- Merged Queries1

Expanded Dim_Admission and added Admission ID to the Billing table.

10. Final Billing Table

Removed unnecessary columns and kept the required fields such as **Admission_ID**, **Billing Amount**, **Date of Admission**, **Discharge Date**, and **Insurance Provider**.



6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Final Billing table with the required columns for analysis.

11. Final Data Model

Created the final healthcare data model by connecting the **Dim_Patient**, **Dim_Admission**, **Billing**, and **healthcare_dataset** tables with relationships.

